Investigating the Adaptive Effects of Ground Compliance in Visual Feedback-Based Propulsion Training

Bradley Hobbs, and Panagiotis Artemiadis *†‡

Abstract

Purpose: This study investigates whether adding ground compliance to visual feedback (VF) gait training is more effective at increasing peak vertical push-off force (POF) compared to using VF alone, with implications for gait rehabilitation.

Design/methodology/approach: Ten healthy participants walked on a custom split-belt treadmill. All participants received real-time visual feedback of their ground reaction forces. One group also experienced changes in ground compliance, while a control group received only visual feedback.

Findings: Intentional increases in vertical push-off ground reaction forces were successfully achieved and sustained post-intervention, especially in the group that experienced ground compliance. This group also demonstrated lasting after-effects in muscle activity and joint kinematics, indicating a more robust learning of natural strategies to increase propulsion.

Originality: This work demonstrates how visual and proprioceptive systems coordinate during gait adaptation. It uniquely shows that combining ground compliance with visual feedback enhances the learning of propulsive forces, supporting the potential use of compliant terrain in long-term rehabilitation targeting deficits in push-off execution and limb loading, such as those following a stroke.

Keywords: robotics, gait, bio-feedback, rehabilitation

Article Classification: Research Article

1 Introduction

In cases of brain lesions, loss of lower limb muscle control is common (Verma et al., 2012), leading to the necessity of complex gait retraining strategies (Anson et al., 2013; Franz et al., 2016; Awad et al., 2014, 2020; Krishnamoorthy et al., 2008; Liu et al., 2021). Increasing a patient's muscle activation and coordination is often a primary goal (Dwyer et al., 2010; Reiman et al., 2012; Akdoğan and Şişman, 2011). It is also clear that joint angle trajectories should be noted during training to ensure proper gait patterns are being developed in training (Vallery et al., 2008; Ji and Manna, 2008; Ensink et al., 2025). Many motor therapy techniques incorporate biofeedback (Huang et al., 2006; Basmajian, 1981; Richards et al., 2017; Lewek et al., 2012; Nelson, 2007) such as live visual feedback (VF) (Sigrist et al., 2013; Giggins et al., 2013) using the participant's own joint angles (Banz et al., 2008) or toe clearance (Banala et al., 2010, 2007) to decrease incidence of drop-foot in participants with hemiplegia (Hsu et al., 2019; Lemoyne et al., 2008; Dingwell et al., 1996).

Force bio-feedback has been used for enhancing anterior-directed push-off forces (Schenck and Kesar, 2017; Genthe et al., 2018; Franz et al., 2014; Herrero et al., 2021; White et al., 1997). These forces are a primary variable of concern (Spencer et al., 2021; Chambers and Artemiadis, 2023; Kim et al., 2024; Roelker et al., 2019; Zelik and Adamczyk, 2016; Hsiao et al., 2015) since gait kinetics can relate mul-

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tiple therapy outcomes (Mirelman et al., 2010; Chen et al., 2007; Hesse et al., 1994; Chen et al., 2003). Indeed, the 3-dimensional push-off force is a conjoining measure of not only the performance of ankle musculature (Kameyama et al., 1990; Schenck et al., 2019), but also in the kinetic chain of the entire lower body as a whole (Dugan and Bhat, 2005), with the anterior-posterior component most responsible for forward-directed propulsive forces (Balasubramanian et al., 2007), and the purely vertical component displaying the largest magnitude (Perry and Burnfield, 1992). Previous work has shown that participants given feedback of only vertical component of ground reaction forces are able to increase POF after training to intentionally increase POF (Hobbs and Artemiadis, 2024).

Modifications to the expected ground reaction force profile during loading response and terminal stance can disrupt sensory pathways to the brain (Perry and Burnfield, 1992; Preusser et al., 2015; Kim et al., 2015). This is one reason investigations over the past decade have included the study of compliant (low-stiffness) environments, which can perturb multiple neural pathways simultaneously. For example, proprioception is altered as the foot descends below the typical ground level during force application, necessitating postural adjustments (Tuthill and Azim, 2018; Rand, 2018; Ting et al., 2015). Inter-limb coordination mechanisms are also challenged when using asymmetric compliant terrain changes (Handelzalts et al., 2019; Skidmore and Artemiadis, 2016b,a), since current research posits that asymmetric gait patterns may be corrected through asymmetric interventions. These multifaceted changes in stiffness aim to stimulate neural plasticity in a unique manner, potentially improving outcome measures on the paretic side.

Crucially, in this study, we hypothesize that the compliant surface functions as a mechanical perturbation that dissipates propulsive energy during the push-off phase (Voloshina et al., 2013; Lejeune et al., 1998). Unlike rigid surfaces, where ground reaction forces are efficiently transferred, compliant terrain absorbs mechanical work, compelling the locomotor system to increase compensatory muscle activity and mechanical power to maintain walking speed (Zamparo et al., 1992; Marigold and Patla, 2002). This added mechanical demand effectively serves as an error-augmentation strategy; by increasing the "cost" of propulsion, the utilization of the compliant surface could stimulate a more robust recalibration of the motor command that persists as a training after-effect (Patton et al., 2006; Reisman et al., 2007).

Compliant environments are typically emulated by walking overground on varied walking surfaces (Xie et al., 2021) similar to deformable ground found in nature. In order to localize and better control the compliance of the walking surface in a laboratory environment, a robotic split-belt treadmill, Variable Stiffness Treadmill (VST), was developed to provide highly repeatable and controllable compliant surfaces (Barkan et al., 2014). Notable findings include significant evoked brain activity through low stiffness perturbations (Skidmore and Artemiadis, 2016b), substantial contralateral muscle activation (Skidmore and Artemiadis, 2016b), and significant after-effects through long-term use (Chambers and Artemiadis, 2023). More recently, VST 2 was developed (Chambers et al., 2025) and is shown in Fig. 1. VST 2 contains numerous improvements to VST, including independent stiffness and speed for both belts, linear vertical deflection instead of angular, wider range of stiffness values, quadrupled walking area and center of pressure sensing area, half of the ground height, and quieter operation (Chambers et al., 2025). At the time of conducting this study, VST 2 is the only treadmill able to adjust the compliance of the walking surface.

This exploratory study aims to evaluate the potential effectiveness of combining reduced ground stiffness with real-time, continuous visual feedback of vertical force for gait training, specifically targeting an increase in push-off force after-effects. We propose a novel protocol and improve upon the current literature in several key ways. First, the present study extends previous work by including compliant terrain during the intervention and studying the associated after-effects. Second, a control group is used to isolate the effects of using visual feedback alone, providing a baseline for comparison. Much of the previous work that has shown promising kinetic results using visual feedback used within-participant study designs. Third, vertical ground reaction forces

are reported, since previous work primarily focuses on anterior-posterior forces during push-off. Fourth, measurements of muscle activity around the hip and knee are reported, as well as joint kinematics while push-off occurs. It is imperative that these are assessed in order to determine fully if visual force bio-feedback can be used to increase push-off force, especially before using this method in conjunction with interventions that utilize more complex forms of robot-assisted therapy (Karakasis et al., 2023).

In summary, this study is based on the hypothesis that integrating low-stiffness compliant terrain with visual feedback (VF) gait training yields greater and more persistent increases in push-off force (POF) compared to training with VF alone. We test this using a novel protocol that evaluates sustained push-off after-effects from this combined approach, offering a comprehensive kinetic and muscular analysis critical for advancing complex robot-assisted gait therapies.

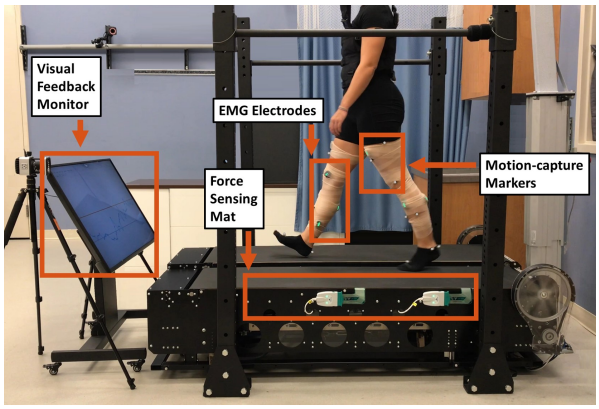


Figure 1: Experimental setup showing a representative participant wearing reflective markers and EMG electrodes on the legs while walking on VST 2 equipped with the force sensing mats, and observing the visual feedback monitor.

2 Methods

2.1 Experimental Setup

For this experiment, a unique instrumented robotic split-belt treadmill, the Variable Stiffness Treadmill 2 (VST 2), was used, as shown in Fig. 1. This device can independently adjust the walking surface stiffness on each belt side via a unique spring-attached lever system. The position of this lever's fulcrum is controlled by a lead-screw-bound carriage that can move with high resolution to different positions, quickly changing the relative stiffness felt by the user. More details on this design are provided in previous work (Chambers et al., 2025). In this study, the left belt was lowered to a stiffness of 25 kN/m, while the right belt remained rigid. Due to the nature of platform height changes, all participants wore a safety harness (Litegait Inc.) capable of supporting the entire body weight of the participant if needed, with the straps adjusted to retain slack even through the lowest platform deflection during a gait cycle with lowered stiffness.

Vertical ground reaction forces for both sides of the VST 2 were captured using four TekScan 3140 Medical Sensors (TekScan Inc.). These flat, mat-like sensors were located under the treadmill belts and, in total, contain 8,448 individual force-sensing cells arranged in a grid pattern, with a resolution of 1 cell per centimeter and a recording frequency of 64 Hz. This allows the center of pressure (COP) location to be precisely determined in real time. The raw data were filtered using a zero-lag fourth-order Butterworth low-pass filter with a cutoff frequency of 20 Hz to attenuate high-frequency sensor noise while preserving the dynamic characteristics of the gait signal (Karakasis et al., 2024). During the experiment, the data were streamed to a computer using software in C# for storing and for utilization in the visual feedback. The visual feedback was displayed on a 90 cm-high-resolution monitor placed directly in front of VST 2 (see Fig. 1). Stored force mat data were synchronized with both EMG and motion capture data through a data acquisition device (National Instruments Corporation) that receives a trigger signal from the Vicon Nexus software (Vicon Motion Systems Ltd). To minimize interference with normal gait mechanics and head position, the monitor was also placed so that the participant's gaze was slightly

downward.

For capturing lower-limb joint kinematics, a motion capture camera system (Vicon Motion Systems Ltd.) was used in conjunction with 20 reflective markers placed at strategic positions. These markers' positions were streamed via 8 high-speed wide-angle cameras at a frequency of 100 Hz. The live marker positions correspond to specific anatomical positions, including the participant's heel, which was used for real-time detection of the heel-strike events that were used to define the start of a gait cycle (Karakasis and Artemiadis, 2021b). Using all 20 markers, the participant's bone positions and orientations were calculated, yielding accurate angle values for all lower-body joints.

Muscle activity was recorded using the Delsys Trigno wireless surface electromyography (EMG) system (Delsys Inc.) at 2000 Hz. A total of 16 electrodes were used, with the recorded muscles corresponding to the following: Tibialis Anterior (TA), Gastrocnemius Medial (GAM), Gastrocnemius Lateral (GAL), Soleus (SOL), Vastus Medialis (VM), Rectus Femoris (RF), Biceps Femoris (BF), and Semitendinosus (ST) for both the left and right legs. Because this is the first study to report vertical ground reaction forces and kinematics using a protocol designed to increase POF, the largest and most active muscles contributing to the movement patterns were included. Temporal synchronization between all data collection devices was achieved through a digital trigger unit and Vicon Nexus 2.12 (Vicon Motion Systems Ltd.). All participants had EMG electrodes placed according to current guidelines (Pinto-Fernández et al., 2024), including shaving excess hair, wiping the skin with alcohol, securing the electrodes to the skin with double-sided tape, and applying an elastic wrap around the limb to reduce movement-induced noise.

2.2 Experimental Protocol

The protocol for this study was approved by the University of Delaware Institutional Review Board (IRB ID#: 1544521-2), with informed consent given by all participants. Participant inclusion criteria include being free from any injuries affecting the lower limbs,

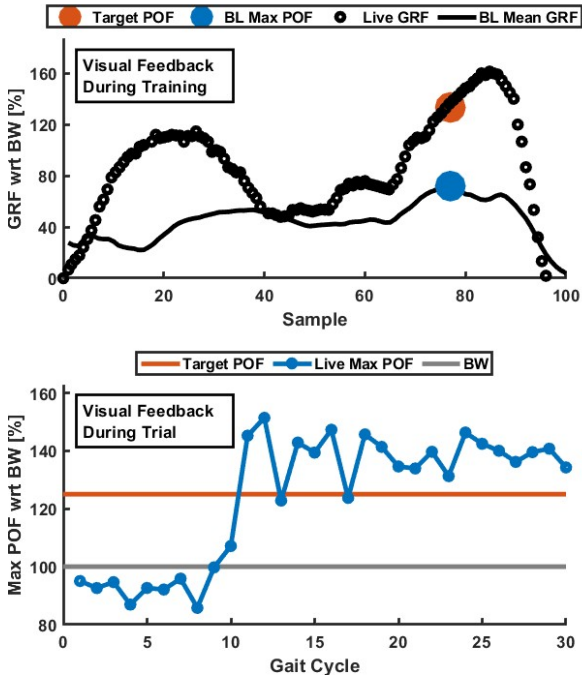


Figure 2: Close look of the visual feedback shown to the participants, with training feedback shown on top and trial feedback shown on the bottom. The axes units, labels, or legends are not shown to the participants, but are included in the figure for clarity, with Baseline (BL) values given with respect to (wrt) percent bodyweight (BW).

gait abnormalities, and attention difficulties. A total of ten participants (5 male, 5 female; age = 25.8 ± 3.1 years; body mass = 72.6 ± 7.9 kg; height = 177.4 ± 9.5 cm) participated in this study, all of whom were right-leg dominant. All participants walked without shoes. The entire experiment was separated into two halves: the Training half and the Trial half.

2.2.1 Training Protocol

For the first half of the study, all participants were instructed to walk normally on VST 2 at a constant speed of 90 cm/s for approximately 100 gait cycles, comprising the Acclimation section of the Training. After the participants were comfortable walking on VST 2, the belts were stopped to go over additional

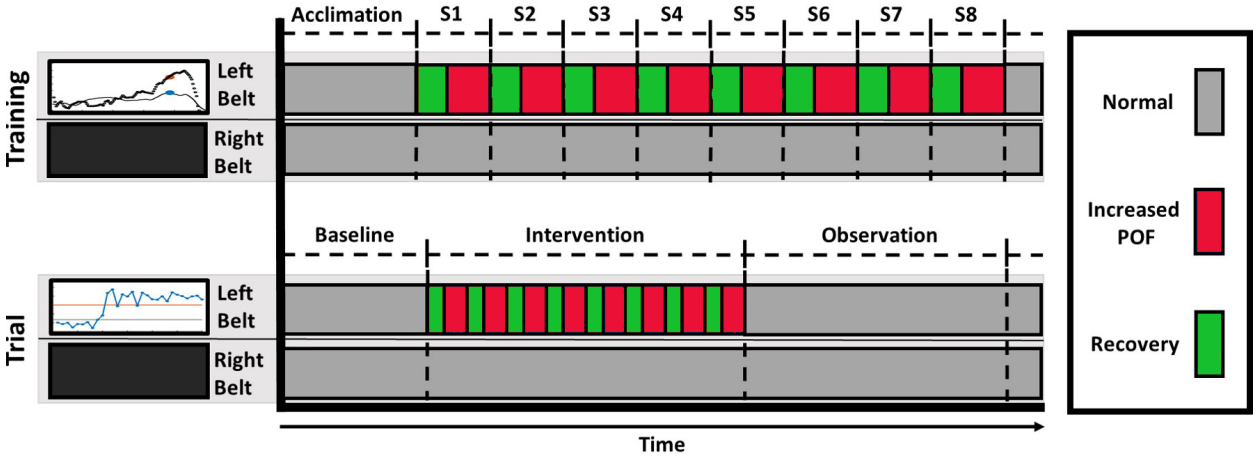


Figure 3: For each group, the visual feedback used is shown on the left, with the relative number of gait cycles in each trial half proportioned through time, and for each belt individually. Each group received the same visual feedback and was equally trained to increase POF.

instructions. Participants were then instructed to closely observe the VST 2 monitor at all times when walking, and that the monitor would provide a visual indication as to the amount of vertical force being exerted on the left belt while walking (see Fig. 2).

For the first ten gait cycles, the participants were shown a blank screen. Then, a baseline average of each participant's ground reaction force (GRF) profile from the left leg only was displayed on screen for the rest of the training half as a static image within a live-updating MATLAB figure window (MathWorks, Inc.). As the participant walked, each ground reaction force magnitude sample was displayed as a black point on the screen, with the x and y coordinates proportional to the current time instant's gait cycle phase and the amount of ground reaction force, respectively. In other words, the participant was able to see the vertical force exerted by the left leg on the treadmill in real time, relative to the calculated baseline over time. A baseline (BL) dot was shown to represent the peak push-off force (POF) for the averaged baseline profile, while a target dot represents 125% of the individual's body weight (BW), since a baseline value was unknown before the start of this study. This value was chosen experimentally and based on previous work (Hobbs and Artemiadis, 2024) to induce a meaningful level of difficulty, while remaining attainable, thereby increasing the participant's motivation without frustration. Setting the target relative to baseline, rather than using an absolute threshold, individualizes the task constraint. This provides a sufficient error-augmentation stimulus to actively drive lower-limb motor adaptation while preventing participants from resorting to atypical, exaggerated gait adjustments (such as running or leaping) to achieve the target. All these are shown in Fig. 2 at the top.

The participants first walked for 10 gait cycles under normal conditions, which constitutes the baseline (BL) phase. This was immediately followed by 15 gait cycles, where participants were instructed to increase the peak force at push-off against the treadmill belt while monitoring their performance on the monitor. This phase is referred to as the increased push-off force (POF) phase. This visual feedback's purpose was to give as much relevant information as possible to allow the participants to adjust the POF strategy in real time. These 25 gait cycles form a single section of the training half, with eight total sections of training (200 gait cycles) given to all participants after acclimation (see Fig. 3). Participants received standardized verbal cues to help them un-

derstand and increase POF, with success defined as reaching the on-screen target in at least 50% of a training section. The target was framed as a binary goal, either reached or not, to encourage consistent effort while minimizing the risk of exaggerated or unnatural gait patterns during the intervention phase.

If a participant didn't succeed within eight training sessions, they rested for up to five minutes before repeating another round, continuing until success was achieved. Most required one or two rounds; two needed three. We ensured all participants received equal training levels before proceeding, followed by a five-minute rest to prevent fatigue.

2.2.2 Trial Protocol

The Trial half of the study consisted of 500 gait cycles separated into three phases: baseline, intervention, and observation (see Fig. 3). The baseline phase consisted of 100 gait cycles with both platforms set to rigid, and served as the participant's normal walking gait for all analysis. The observation phase operated under the same conditions as the baseline, but lasted for 200 gait cycles in order to give ample time to observe after-effects from the intervention phase. The purpose of this phase was to be identical to the baseline in protocol, so that changes lasting from the intervention phase could be measured against those before the intervention. The intervention phase immediately follows the baseline phase and occurs before the observation. In the intervention phase, participants walk for 200 gait cycles, completing exactly one additional round of eight sections, structured exactly as in training (see Fig. 3). Visual feedback targets were adjusted for all participants to align with the specific objectives of each experimental phase.

The visual feedback used in the Trial intervention phase for all participants was a simpler version of the training visuals, because the participants were already familiar with the task, and did not need as much detailed information. As shown in Fig. 2 (bottom), a target was displayed as a solid line across the screen. Another line was shown representing the individual's body weight to provide a ground reference for the scale. The live feedback representation shown in this visual, however, was only the POF of the current gait cycle, which was calculated and displayed back within five milliseconds (ms) after the left toe-off (LTO) occurred. Specifically, the participant was shown a single new point for each gait cycle, providing POF performance feedback relative to the target at the end of each stance phase. This point remained on the screen for several gait cycles, allowing the participant to visually gauge effort relative to previous cycles, a feature that has never been used in this type of intervention protocol.

It was during the intervention phase that the participants were randomly divided into two groups: the Visual Feedback with Low-Stiffness (VF-S) group, and the Visual Feedback with rigid (VF-R) group, serving as a control. The VF-S group had the left platform of VST 2 set to reduced stiffness (25 kN/m) only during the period when POF was intentionally increased (15 gait cycles), then set back to rigid for the next 10 gait cycles. Because this low-stiffness environment had not been experienced by the participants previously, the VF-S group had to adapt to this change in real-time. All participants still had eight attempts to reach the target shown on the screen, with the VF-S group requiring low-stiffness walking on the left side. In order to duplicate any auditory feedback experienced by the VF-S group due to the stiffness changes, the VF-R group protocol includes having the VSM move out during the swing phase, but return to rigid values before heel-strike. It is noted that participant blinding was unfeasible due to the discernible physical feedback of the compliant substrate; however, investigators were blinded during initial data preprocessing. Independent t-tests confirmed baseline equivalence between groups, with no significant pre-intervention differences observed in body weight or baseline peak push-off force ($p > 0.05$).

2.3 Data Pooling and Processing

Using the same method for each participant, all data processing was completed individually, with further separation between participant groups, experiment halves, and walking phases. Then, for each participant individually, data outliers were removed using a systematic trajectory-based algorithm (Hobbs and

Artemiadis, 2022). Specifically, gait cycles were isolated, and an ensemble average was calculated. Cycles exhibiting a Euclidean distance greater than two standard deviations from the median trajectory at any point in the gait cycle were flagged as outliers and excluded from further analysis to ensure data integrity.

For analysis of POF across the trial half, all participants were pooled to draw conclusions about the intervention as a whole regarding POF, which was the primary focus of the present work. For secondary analysis, two participants were selected from each group to represent each group's average, with an equal number of participants performing higher and lower on POF. This allows direct comparison between groups without normalizing further data to an average.

Statistical analysis was performed using Linear Mixed-Effects (LME) models (MATLAB Statistics and Machine Learning Toolbox) with $\alpha = 0.05$. This approach was selected to explicitly account for the hierarchical structure of the data, where multiple steps are nested within individual subjects, thereby correcting for within-subject correlations that standard pooling methods neglect. To formally evaluate group-dependent differences in retention, the fixed-effects structure of the primary model included Group (VF-S vs. VF-R), Experimental Phase (Baseline vs. Observation halves), and their interaction term (Group $\times$ Phase). Subject-specific variability was accounted for by including 'Subject' as a random effect (random intercept). The primary hypothesis of a group-dependent retention effect was evaluated via the significance of the Group $\times$ Phase interaction coefficient. Finally, the unit of analysis for the primary LME models was the individual gait cycle (step), thereby maintaining high-resolution variance tracking.

Following the evaluation of the global interaction effect, pairwise comparisons displayed in the figures (e.g., within-group differences between baseline and observation phases) were derived as planned post-hoc contrasts calculated directly from the LME model structure. These simple effects were evaluated by testing linear combinations of the model coefficients, ensuring that all within-group comparisons maintained the appropriate corrections for subject-level clustering and nested dependencies. False discovery rate (FDR) corrections were applied to adjust for multiple post-hoc comparisons.

For visualization purposes, the POF for the Trial phase was compiled for each group, normalized to each participant's body weight, and expressed relative to the average POF of their respective baseline phase. Linear Mixed-Effects models were employed to determine if the POF in the VF-S group was significantly higher than that of the VF-R group, accounting for subject-specific variability. Ground Reaction Force (GRF) data were normalized to body weight and filtered following methods established in previous work (Hobbs and Artemiadis, 2024). Both left and right GRF profiles are presented for the stance phase only; the left leg data spans from left heel-strike (LHS) to left toe-off (LTO), while the right leg data spans from right heel-strike (RHS) to right toe-off (RTO).

In the visualizations, statistical significance is denoted by an asterisk connecting the datasets. For POF data, all gait cycles in the baseline phase were included, while the observation phase was split into two halves ('Early' (H1) and 'Late' (H2)) and evaluated separately to capture the transient nature of the adaptation. Each half of the observation phase was compared against the entire baseline phase. For boxplots, the specific comparison is indicated at the top of the significance line. The directionality of the effect, determined by the sign of the model's estimated coefficient, is indicated by arrow symbols: an upward arrow indicates a significant positive effect (Observation $>$ Baseline), while a downward arrow indicates a significant negative effect (Observation $<$ Baseline). The absence of an arrow indicates a non-significant result.

While primary confirmatory hypotheses focused strictly on the vertical push-off force (POF) metric, the extensive family of secondary EMG and kinematic metrics was designated as exploratory. Consequently, uncorrected alpha levels were used for these secondary metrics to identify broad physiological adaptation trends and to generate hypotheses for subsequent full-scale trials.

Motion capture data were used to calculate 3D

joint angles using the Vicon Plug-in-Gait model (Vicon Motion Systems Ltd.), which uses 3D marker positions and each participant's anatomical parameters. EMG data were filtered with a fourth-order Butterworth band-pass filter with cut-off frequencies at 30 and 300 Hz. After full-wave rectification, a 200-point moving average was used to calculate the muscle activity envelope, and a fourth-order, 5 Hz lowpass filter was used. Data were then normalized to the highest recorded value during the experiment, the experiment maximum (EM). This normalization method consists of the peak dynamic method (Burden, 2010), which reduces inter-participant variability when individual muscle effort comparisons are not needed. For both EMG and kinematic data, dynamic time warping was performed using a standard spline to the highest number of samples available in a given participant, and spans between left heel-strikes, which were calculated offline using the heel marker (Karakasis and Artemiadis, 2021a).

To characterize generalizable after-effects, data from all 10 participants were included in the group-level LME analysis. In addition to the group-level statistics, detailed time-series data are presented for a representative participant from each group. This allows for the visualization of intervention effects on absolute force magnitudes without the potential masking effects of normalization. These representative participants were selected to illustrate typical group performance, identified by their proximity to the group mean rather than as high-performing outliers.

Throughout the results, data for the VF-S group and VF-R group are consistently displayed in blue and orange, respectively. Finally, this study focuses on the observation phase compared to baseline; the intervention phase is not analyzed here, as it was previously studied using visual biofeedback during rigid walking (Hobbs and Artemiadis, 2024).

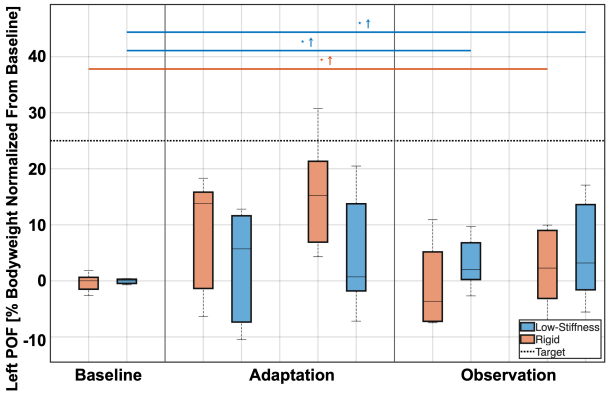


Figure 4: Group-level analysis of peak vertical Push-Off Force (POF) values. The Adaptation and Observation phases are subdivided into First Half (H1) and Second Half (H2) to illustrate the temporal progression of learning and retention. Data are normalized to the average POF of the Baseline (BL) phase for each participant. Blue bars represent the Low-Stiffness group (VF-S), and Orange bars represent the Rigid group (VF-R). Asterisks ($*$) and connecting lines indicate statistically significant within-group differences ($p < 0.05$) between the specific phase half and the Baseline, derived as planned post-hoc contrasts (simple effects) from the primary Linear Mixed-Effects (LME) model conditional on the Group $\times$ Phase interaction effect. Error bars represent the standard error of the mean.

3 Results

3.1 Intervention (Adaptation) Phase

All participants met or exceeded the POF target in at least 50% of the final training section and retained increased POF into the adaptation phase, often surpassing prior maximums. This was accompanied by a 14% increase in stance-phase duration, and all groups consistently exceeded the 125% bodyweight POF target, aligning with previous findings (Hobbs and Artemiadis, 2024).

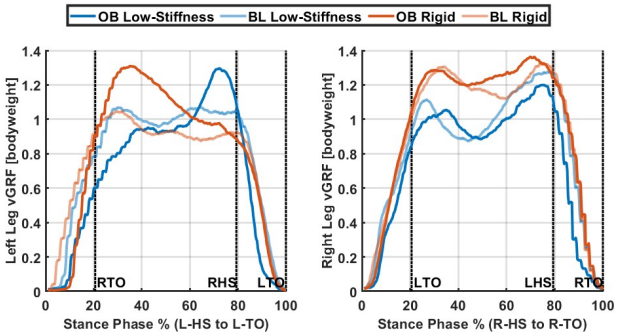


Figure 5: Ground Reaction Force (GRF) profiles for a representative participant from each group. Blue lines correspond to the Low-Stiffness (VF-S) participant, and Orange lines correspond to the Rigid (VF-R) participant. Lighter shades represent the Baseline (BL) phase, while darker shades represent the Observation (OB) phase. Dashed lines indicate the transition between stance and swing phases. Force values are normalized to Body Weight (BW).

3.2 Observation Phase vs. Baseline Phase

The LME analysis revealed a significant difference between groups ($p < 0.05$), indicating that participants in the low-stiffness condition (VF-S) retained a significantly larger increase in POF compared to the rigid condition (VF-R). While the limited sample size warrants a degree of caution, these statistical results provide clear preliminary evidence of a group-dependent adaptation response. Notably, the VF-S group sustained this elevated POF throughout the entire observation phase, with no marked decay towards the end of the trial (see Fig. 4). In contrast, participants in the VF-R group exhibited a delayed and attenuated response, where a minor increase in POF appeared only in the latter half of the observation phase. It should be noted that while propulsive strategies involve multi-axial forces, the target metric analyzed here is strictly the peak vertical component of the ground reaction force during the push-off phase.

Analysis of the ground reaction force (GRF) profiles for the VF-S group revealed an approximate 20% increase in left leg stance-phase duration, which coincided with the significant increase in left push-off force (POF) from baseline, as shown in Fig. 5 for a set of representative subjects per group. In contrast, no significant changes were observed in the right leg GRF profile for this group.

To gain insight into the underlying motor adaptations driving push-off alterations, an exploratory analysis of lower-limb muscle activity and joint kinematics was conducted. In the text below, continuous joint-angle trajectories and EMG amplitude differences refer to trends observed in representative subjects.

Just after the left heel-strike, a knee flexion increase of 2% was seen in the left leg of the VF-S group, while the VF-R group saw a decrease of 5% (see Fig. 6). At the same time, the left ankle had an increase in peak ankle dorsiflexion of 2% at heel-strike for the VF-S group, while the VF-R group had a decrease of 5%. These changes were also coupled with a significant change in left rectus femoris activity, with a 45% increase for the VF-S group, and a 30% decrease in the VF-R group (see Fig. 7).

Just before peak POF in the observation phase, many changes relative to baseline were observed in both groups. The left hip angle has a small 3% increase for the VF-S group, while the VF-R group has a 3% decrease. Both changes occur from heel-strike until POF initiation. For the VF-R group, this was coupled with a decrease in lateral gastrocnemius activity of 20% through the stance phase leading up to peak POF, with the VF-S group maintaining the same level of activity compared to baseline.

Changes to the contralateral leg were also observed. The VF-S group does have an increase of 10% for the right biceps femoris at POF initiation, while the VF-R group has a rather large decrease in peak activation of 41%. For the right leg, the VF-S group has an increase in ankle plantarflexion by 9%, while the VF-R group shows no change. At the same time, subsequent changes were observed in right knee extension, with the VF-S group showing a small increase of 5% and the VF-R group showing a small decrease of 4%.

Distinct biomechanical adaptations were observed immediately after participants achieved peak POF

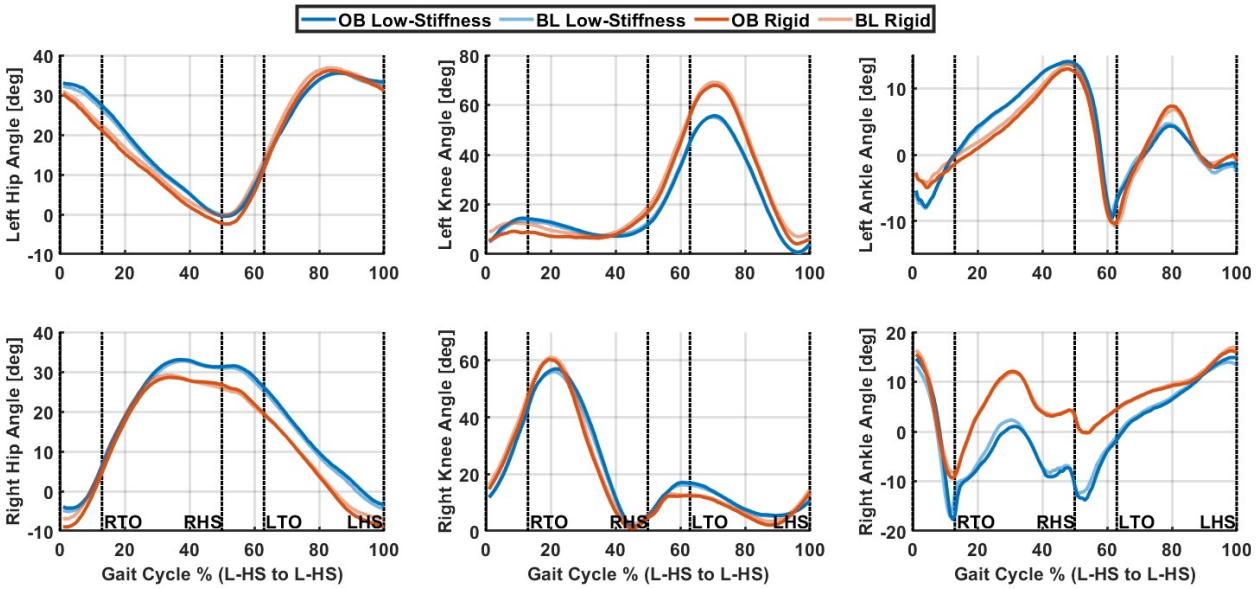


Figure 6: Joint kinematics (left and right legs) for representative participants selected to illustrate the group mean behavior. Blue lines correspond to the Low-Stiffness (VF-S) participant, and Orange lines correspond to the Rigid (VF-R) participant. Lighter shades represent the Baseline (BL) phase, while darker shades represent the Observation (OB) phase. Dashed lines indicate the transition between stance and swing phases.

during the observation phase. For the left leg, a small increase of 2% was seen in knee flexion of the VF-S group, while the VF-R group at the same time frame had a 5% decrease. The left leg also had a decrease of 9% in peak hip extension for the VF-R group, with the VF-S group seeing sustained peak hip extension.

For the right leg, an increase in the second peak of ankle plantarflexion of 9% during maximum POF for the VF-S group is observed, while the VF-R group has no increase relative to baseline. Muscle activity for the right leg during this period shows an increase of 9% for the VF-S group biceps femoris, while the VF-R group has a 19% decrease (see Fig. 7). This was coupled with a 22% increase in the rectus femoris of the right leg in the VF-S group, with only a 3% increase in the VF-R group. At the same time as these changes, the right tibialis anterior has a significant increase of 18% during the right swing phase of only the VF-S group, and the VF-R group decreases activity during this period by 13%. For the remaining muscles recorded, no significant changes in activation amplitude were observed between phases or groups.

Immediately following left toe-off of the observation phase, the VF-R group has a small decrease in peak knee flexion of 2-5%, while the VF-S group maintains peak knee flexion relative to baseline. Coupled with this change were several notable changes in left leg muscle activity. The vastus medialis decreases by 35% in the VF-R group, and the VF-S group maintains muscle activity. Both of the hamstring muscles have an average increase of 30% for the VF-S group, while the VF-R group decreases by 29% in average hamstring muscle activity. Also notable is that the left tibialis anterior has an increase of 12% for each peak activity during the left swing phase for the VF-S group, with the VF-R group decreasing activity by 40%. The right leg for the VF-S group has a small increase of 4% for peak hip extension and 2-6% for

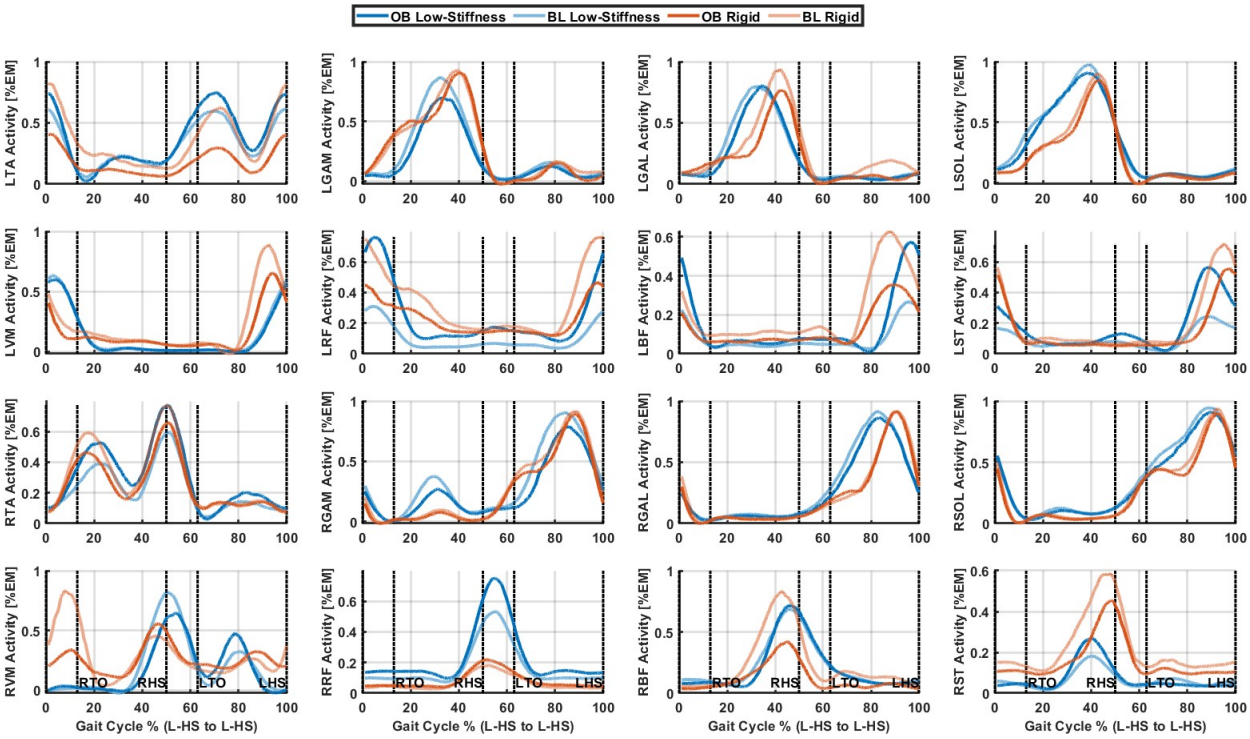


Figure 7: Resulting muscle activity recorded for both left and right legs for representative participants from each group. Blue lines correspond to the Low-Stiffness (VF-S) participant, and Orange lines correspond to the Rigid (VF-R) participant. Lighter shades represent the Baseline (BL) phase, while darker shades represent the Observation (OB) phase. The units are normalized to the experiment maximum (EM).

the VF-R group.

4 Discussion

4.1 Ground Reaction Force Adaptations and Bilateral Loading

The stance-phase duration increase of the left leg correlates with previous work (Hobbs and Artemiadis, 2024). This finding is a direct side effect of the strategies used to intentionally increase POF. For analysis of the GRF profile, the most notable change in the VF-S group occurs with the POF having increased from baseline on the left leg (see Fig. 5). This change aligns with the rest of the VF-S group, with some participants showing larger after-effect increases in POF and others showing smaller increases. Also noted was a lower peak weight-acceptance force, shifted later in the gait cycle, which was not observed in the VF-R group. In fact, the VF-R group had an increase in weight acceptance force with a magnitude similar to the increase in peak POF seen in the VF-S group. Both groups had very similar peak weight acceptance forces for the baseline phase of the left leg. This observed adaptation, along with the expected increase in stance-phase duration, a known correlate of intentional POF increase (Hobbs and Artemiadis, 2024), suggests that the compliant environment facilitates a more natural and mechanically efficient gait strategy. This ability to decouple propulsive gains from

high-impact landings is a key finding that supports the therapeutic potential of compliant-surface training. However, the observation that the VF-R group exhibited a small increase in peak POF during the intervention suggests that the VF-S condition may make it more difficult to increase POF during an intervention phase. Further work analyzing effort for each of these conditions could help elucidate whether the VF-S environment requires more effort to increase POF, whether the effect was due to specific strategies used, or whether it was due to the nature of the force-mat sensor.

While previous studies have successfully targeted POF increases using visual biofeedback alone (Genthe et al., 2018; Schenck and Kesar, 2017) or fast-walking interventions (Awad et al., 2014), these methods often require significant conscious effort and may not always transfer effectively to overground walking without the feedback stimulus. Our results suggest that incorporating the VF-S condition may offer a distinct advantage over these purely visual or kinematic approaches. By altering the mechanical environment to reward forceful push-offs with a stiffer, more stable response (as the spring compresses), the VF-S condition may implicitly drive the nervous system toward a more robust propulsive strategy, potentially reducing the cognitive load required to maintain high POF compared to visual targets alone.

Furthermore, compared to robotic interventions that directly apply assistive forces to the ankle (e.g., exoskeletons; (Mooney et al., 2014)) or functional electrical stimulation (Hakansson et al., 2011), the VF-S paradigm encourages the user to voluntarily generate force against a compliant substrate. This active engagement is critical for motor learning. While calculating the direct efficacy of VF-S against these other established modalities is beyond the scope of this single study, our data regarding the retention of POF in the VF-S group aligns with principles of error-augmentation learning, suggesting that environmental manipulations may be as effective, if not more so, for retention than guidance-based therapies.

For the right leg, vertical GRF modestly decreased in the VF-S group and slightly increased in the VF-R group, suggesting a potential compensatory weight shift resulting from the intervention. This shift may explain the observed increases in left leg GRF and points to a meaningful after-effect: overall weight redistribution. Given that stroke survivors often exhibit reduced vertical GRF and propulsive forces on the paretic leg (Bowden et al., 2006; Balasubramanian et al., 2007), these results are especially relevant. They highlight the potential of this approach to promote more balanced loading and improve gait symmetry, a key metric often correlated with functional recovery in stroke rehabilitation (Patterson et al., 2008).

4.2 Stance Phase Strategies: Joint Stiffness and Muscle Activation

Findings from the preparation leading up to increasing POF indicate that participants who received low-stiffness training lowered their center of mass by decreasing specific joint angles, providing evidence that the span between the hip and ankle was reduced. This finding is notable, as it indicates, for the first time, that joint stiffness was reduced as an after-effect of only low-stiffness POF training, and was likely a rebound effect of having to increase joint stiffness to overcome the compliant terrain. Also corroborating these results was the marked decrease in the peak weight acceptance force for the VF-S group, further strengthening this claim.

The VF-S group exhibited a substantial increase in rectus femoris (RF) activity, which coincided with notably greater knee flexion during the loading response. Given the RF's biarticular role in knee extension and hip flexion, this elevated activity likely serves a dual purpose: providing dynamic knee stability to manage the compliant surface and controlling the forward torso lean required for higher propulsion. This finding may have important translational implications for stroke rehabilitation. Many stroke survivors develop pathological joint stiffness, such as a "stiff-knee gait," as a maladaptive compensatory strategy for muscle weakness and instability (Li et al., 2021). This excessive rigidity is detrimental: it severely limits shock absorption, prevents a normal loading response, and blocks the kinematic adjustments needed for efficient propulsion. Our interven-

tion appears to directly challenge this pathological pattern. By promoting increased knee flexion, the compliant surface discouraged a simple, rigid stabilization strategy. The corresponding surge in RF activity is therefore not pathological stiffening; rather, it represents a functional, learned muscular response to actively control a more dynamic and flexible joint. This is a critical distinction, suggesting a shift from a passive, locked-joint strategy to an active, controlled one. If this after-effect translates to paretic gait, it represents a potential method for re-training a more normative loading response, a crucial prerequisite for restoring shock absorption and improving propulsive power.

During push-off, the left leg kinematic changes were small, but sustained throughout the stance phase. It cannot be concluded that the early increases in rectus femoris seen in this period were intended to increase hip flexion in the group that underwent the compliant terrain. Rather, the small increase in hip flexion further supports the conclusion that knee joint stiffness was the likely target. The relative decrease in lateral gastrocnemius activity in the VF-R group compared to the VF-S group suggests that the present intervention for this muscle was critical, since the gastrocnemius plays a large role during the intervention phase's increased POF strategy. It is apparent that decreased activity compared to baseline is seen as an effect of intentionally increasing protocols in the rigid case; however, this effect seems to disappear with the addition of lowered stiffness during the intervention.

It is also worth noting the specificity of the observed muscle adaptations. Significant retention was observed primarily in the propulsive agonist muscles (e.g., Gastrocnemius, Biceps Femoris), rather than a global increase in activation across all recorded muscles. While these results are preliminary, this selectivity may suggest that the VF-S condition facilitated a task-specific motor learning effect, where participants optimized the recruitment of prime movers without resorting to inefficient, generalized co-contraction strategies often seen in early stages of gait rehabilitation. Further investigation is needed to confirm whether this specific activation pattern allows for more efficient gait in the long term compared to global muscle recruitment.

4.3 Swing Phase Mechanics and Bilateral Coordination

At peak push-off during the observation phase, adding compliant terrain while increasing POF in the intervention phase mitigated the decrease in peak hamstring activation of the contralateral (right) leg and increased ankle plantarflexion. Based on these results of the right leg, the further extended knee and pointed foot indicate that one potential after-effect of this study is an increase in target step length of the opposing leg. This finding aligns with known interventional strategies to address asymmetric step length in post-stroke individuals. At peak push off, the paretic leg has a marked decrease in POF, with the contralateral leg having a corresponding decrease in step length. An increase in target step length directly counteracts the deficiency during training to increase POF. This effect was observed only in the VF-S group. Further research is needed to determine whether overall step length asymmetry can be reduced with this intervention.

While small, the increase in knee flexion after push-off at the initiation of the swing phase is a promising finding, as it may have implications for increased toe clearance as an after-effect of the intervention. A decrease in toe clearance was observed in individuals post-stroke, so further research is needed to determine whether an increase in toe clearance adds value to this protocol. While both groups have an observed increase in POF compared to baseline, the decrease in POF for the VF-R group could be partially explained by the decrease in peak hip extension following peak push-off, connecting both the GRF and POF kinetic results with the kinematic observations.

In particular, the large increase in right tibialis anterior activity is important because increased tibialis anterior activation during the swing phase is a target of many interventions aimed at decreasing the presence of drop-foot and increasing toe clearance. Walking under unilaterally compliant terrain has been shown to increase tibialis anterior activity of the contralateral leg during the intervention (Skidmore and Artemiadis, 2016a, 2015). This study

shows, for the first time, that adding an intentionally increasing POF session, coupled with unilaterally low-stiffness walking, not only increases contralateral tibialis anterior activity but also retains this effect as an after-effect of the intervention. Because this increase in activity has been shown in patients post-stroke (Skidmore and Artemiadis, 2017), this finding provides further evidence that this protocol has merit for translation to gait rehabilitation post-stroke, using the VST 2.

The finding of left tibialis anterior activity increase in the targeted side for the intervention is promising, since it may indicate that not only can ipsilateral stance phase kinetics be increased through adding low stiffness walking, but potentially swing phase toe clearance as well, broadening the scope of such a protocol to other pathologies known in the gait of individuals post-stroke. Combined together, the results for the VF-S group show a synergistic relationship between the left tibialis anterior and both left hamstrings in maintaining leg clearance during the left swing phase. Because muscle activity is significantly altered from baseline in the VF-S group, yet the increase in peak knee flexion is relatively minor, further research is needed to understand the potential outcomes of this increase in muscle activity.

4.4 Study Limitations and Future Work

Several limitations of this study must be acknowledged. First, the sample size (N=5 per group) is limited, which restricts the generalizability of the findings. To address this, we employed LME models, which maximize statistical power by utilizing the full resolution of step-level data rather than reducing participants to single averages. Despite this rigorous statistical approach, the small cohort size means these results should be interpreted as preliminary evidence requiring validation in larger populations. Future studies may utilize these effect estimates to perform power analyses for larger-scale trials.

Another limitation of this work is its reliance on a healthy cohort, leaving the translation of these outcomes to clinical populations an open question. While our specific findings on visual-compliance training are novel, the research pathway from healthy participants to patients is well-defined. Studies on split-belt adaptation, for example, have consistently shown that results from healthy cohorts can translate successfully to patient populations (Reisman et al., 2005, 2007). Encouragingly, some studies have even found greater rehabilitative effects in patients than were observed in healthy participants (Reisman et al., 2009; Herrin et al., 2024). Consequently, the clear future direction is to apply the present protocol to target populations (e.g., stroke survivors) to investigate its therapeutic potential.

Another limitation of this study is that our kinetic analysis was restricted to the vertical component of the ground reaction force (GRF). Due to constraints in our force-sensing equipment, we did not capture the anterior/posterior (A/P) forces, which are critical for a comprehensive understanding of walking kinetics and propulsion. Future work must utilize 3-axis load cells to measure the complete GRF profile. This enhanced instrumentation will allow direct analysis of propulsive A/P forces and enable tracking of the center of pressure, providing deeper insight into strategies to increase POF.

Moreover, this study did not directly quantify knee joint stiffness, representing a key limitation. Our interpretations are based on indirect muscle activity patterns, such as the observed decrease in activity for both knee flexors and extensors (e.g., vastus medialis) during the swing phase in the rigid group. This preliminary finding suggests a potential decrease in swing-phase stiffness for that group. This is notable as it contrasts with the increase in stance-phase stiffness inferred for the VF-R group. These opposing, and currently speculative, findings highlight the need for future work to use inverse dynamics to directly calculate joint stiffness. This will be essential to confirm and understand how these two training methods differentially modulate stiffness across the gait cycle.

Finally, beyond technical constraints, we must consider the clinical feasibility of the VF-S training strategy. Walking on compliant surfaces is known to increase metabolic cost compared to rigid ground (Lejeune et al., 1998), which could induce earlier fatigue in patient populations with reduced endurance, such as stroke survivors. Consequently, future clinical proto-

cols may need to adjust training duration or provide rest intervals to manage this increased effort. Regarding safety, the instability introduced by the VF-S condition, while beneficial for challenging stability and promoting adaptation, poses a potential fall risk. Therefore, clinical implementation would likely require safety harnesses or body-weight support systems, as those used in our study. Despite these physical challenges, we hypothesize that adherence could remain high; the gamified nature of the visual feedback, combined with the distinct mechanical feedback of the VF-S surface, may offer a more engaging environment than repetitive overground walking.

5 Conclusion

This exploratory study investigated the after-effects of an intervention using visual force biofeedback to increase push-off force (POF) while walking in a compliant environment. Our preliminary results suggest that including low-stiffness terrain as a training modality may enhance retention of gait outcome measures compared with walking on rigid terrain. Specifically, we observed that walking on unilateral low-stiffness environments, such as the one emulated by the VST-2, was associated with lasting increases in the activity of several muscles and joint angles during key moments in the gait cycle. These findings support the potential utility of using compliant surfaces as a mechanical perturbation to augment motor learning. However, given the limited sample size, these results should be viewed as a proof of concept. Future work should focus on validating these effects in larger cohorts, modeling direct joint kinetics, and explicitly studying clinical populations (e.g., post-stroke) to determine if these strategies translate to functional rehabilitation benefits.

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